

Ashkan Aghamoali

Mechanical Intelligence Research Group, London South Bank University

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I build **extended-reality applications and digital twins** for engineering – AR interfaces that put simulation results, neural-network predictions and robot state into the room with the engineer, and simulation-first twins running the full stack from an STM32 edge node up to an operations dashboard. Six years with a **bio-inspired structural engineering** group in London, where the XR work began.

EXPERIENCE

AI Researcher / XR Developer

2020 – present

Mechanical Intelligence Research Group, London South Bank University – London, UK

- Develop bio-inspired structures within a multidisciplinary group spanning biology, mechanics and computation.
- Built a Python automation system integrated with CAE software for parametric modelling, and analysed over 1,000 Abaqus simulation instances through it.
- Trained machine-learning models of structural behaviour on that corpus, with optimisation and sensitivity analysis feeding back into design tuning.
- Developed the group's XR software for iOS and Android, presenting products, simulations and research findings in AR.

Research Assistant — computer vision

2016 – 2020

University of Guilan – Rasht, Iran

- Built a detection system for camouflaged moving targets in video; a hybrid HSV and Local Binary Patterns approach improved accuracy by 40%.
- Wrote custom CUDA kernels for an 8× speed-up, enabling real-time 4K analysis at 30 fps, and an OpenGL visualisation module with dynamic thresholding.

EXTENDED REALITY

Six AR applications built at LSBU. They are screen recordings rather than screenshots, and they play on the portfolio: iamashkan.github.io/portfolio

Neural network prediction interface

Unity · ML · C#

An in-scene keypad takes design parameters and returns neural-network predictions of structural deformation as live metrics and charts beside the model.

Parametric control panel

Unity · ARKit

Real-time tuning of a lattice — density, radius, height, thickness — with the geometry rebuilding as the values move.

Bio-inspired structure in AR

Unity · Biomimetics

A biomimetic structure with live controls to morph its geometry and step through design variations.

Step-by-step AR explanation

Unity · 3D

An immersive walkthrough of a bio-inspired structure and its simulation results, anchored in the viewer's own room.

AR co-bot interface

Unity · Robotics

An interface for collaborative-robot visualisation, linking digital simulation to physical automation.

A robotic joint rendered live with sensor and data overlays, bridging simulation data and physical motion.

DIGITAL TWIN SYSTEMS

Five simulation-first systems, each running the whole stack — edge node, decision service, control twin, operator dashboard. Source at github.com/iamashkan

Real-time digital twin generation on iOS

An iPhone scans a space into a live, queryable 3D twin; on-device vision and nameplate OCR bind each asset to the real equipment.
Swift · ARKit · RoomPlan · Core ML · MQTT IN DEVELOPMENT

UR5e circular motor recovery cell

A cobot sorts eleven components of an end-of-life motor into reuse, repair, replace or recycle, scored from its product passport.
ROS 2 · MoveIt 2 · Gazebo · Unity PERCEPTION MOCKED

Battery sorting digital twin

Vision and NASA state-of-health data route each end-of-life cell to reuse, remanufacture, recycle or quarantine under the EU battery passport.
Unity · FastAPI · scikit-learn SIMULATION ONLY

Emergency-aware highway lighting

Six lanes lit only where traffic is, escalating to a full safety corridor ahead of a detected emergency vehicle.
STM32 · MQTT · Unity 6 · Streamlit EDGE TO CLOUD

Defect inspection twin

Surface defects detected and localised with Grad-CAM, then graded into reuse, repair or recycle inside a robotic inspection cell.
PyTorch · Grad-CAM · ROS 2 · Gazebo SIMULATION ONLY

RESEARCH

S. H. Eraghi, A. Toofani, **A. Aghamoali**, A. Zoroufi, M. Shafaghi, S. Jafari, S. Goel, H. Rajabi. "The rise of metaverse manufacturing: towards a universal, sustainable and bioinspired industrial ecosystem." *Journal of the Royal Society Interface*, vol. 23, 2026. doi:10.1098/rsif.2025.0480

A. Toofani, S. H. Eraghi, **A. Aghamoali**, A. Zoroufi, A. Khaheshi, A. Basti, S. Gorb, H. Rajabi. "One to many: an intelligent biomimetic design framework for functional diversification." In review, 2025.

A. Aghamoali, M. Nahvi. "Improvement of camouflage moving target detection in video sequences." *Journal of Mathematical Modeling*, 2020.

Gold Medal — 3rd International Invention & Innovation Competition, International Federation of Inventors' Associations (IFIA), August 2023.

TECHNICAL

EXTENDED REALITY	Unity 3D, ARKit, RoomPlan, RealityKit, WebXR, Needle Engine, Shader Graph, C#, Swift
DIGITAL TWINS	ROS 2, MoveIt 2, Gazebo, MQTT, STM32 and embedded C++, RS485 / CAN, FastAPI, PostgreSQL
MACHINE LEARNING	PyTorch, scikit-learn, computer vision, object detection, Grad-CAM, surrogate modelling, CUDA
ENGINEERING & PLATFORM	Abaqus / CAE automation, parametric and optimisation workflows, Python, TypeScript, Node.js, Docker

CERTIFICATION

Machine Learning and Deep Learning Specializations
Stanford / DeepLearning.AI · Deep Neural Networks with PyTorch IBM · AR/VR/MR/XR with WebXR, Unity & Unreal and Digital Technologies & the Future of Manufacturing Michigan · Mobile AR Development, Unity Essentials Unity

EDUCATION

MSc, Digital Electronic Systems
University of Guilan, Iran · 2016 – 2019

BSc, Electrical & Electronics Engineering
University of Guilan, Iran · 2012 – 2016